

MEDAXONUM

Name and dosage form of product	MEDAXONUM 1G POWDER FOR SOLUTION FOR INJECTION/INFUSION
Therapeutic class	Long-acting broad spectrum 3 rd generation cephalosporin antibiotic
Description	An almost white or yellowish crystalline powder, slightly hygroscopic <i>After reconstitution: clear, yellowish solution</i> <i>List of compatible diluents:</i> <i>For IM injection: 1.0% Lidocaine.</i> <i>For IV injection: water for injection.</i> <i>For IV-Infusion:</i> 1. Dextrose Injection BP 5% 2. Sodium Chloride Injection BP 3. Sodium Chloride and Dextrose Injection BP 4. Dextran 6% in Dextrose Injection BP 5%, 5. Hydroxyethyl Starch 6% infusions <i>Diluent will not be provided with this product</i>
Name and strength of active ingredient	MEDAXONUM 1g vials contain 1g ceftriaxone as ceftriaxone sodium
Pharmacology	Pharmacokinetics Ceftriaxone exhibits concentration dependent binding to serum albumin proteins, and it is this factor that mainly determines the pharmacokinetics. Over the majority of the therapeutic concentration range the unbound fraction of ceftriaxone is approximately 5%, rising to about 15% at a concentration of 300mg/L. Concentrations of free ceftriaxone in the interstitial fluid are correspondingly higher than plasma concentrations due to the lower albumin content of interstitial fluid. Intramuscular administration of 500mg ceftriaxone results in peak plasma concentrations of 40mg – 70mg/L within 60 minutes, bioavailability following administration is 100%. Intravenous injection results in peak plasma levels of approximately 120mg/L after a 500mg dose, and 200mg/L following a 1g dose. A 30 minute infusion of 2g gives a plasma level of about 250mg/L. The primary excretion route is via the kidney, virtually all as a result of glomerular filtration, and approximately 60% of the dose is excreted in the urine as unchanged ceftriaxone. The remaining fraction of the dose is excreted via the biliary and intestinal tracts. Renal clearance is reported as 5ml – 12ml/minute, with reported total plasma clearance rate of 10ml – 22ml/minute. Adult elimination half-life is approximately 8 hours, dose, administration route, and repeat dosing do not significantly affect the half-life. In neonates, approximately 80% of a dose is excreted in the urine during the first 7 days, the proportion declines to levels similar to adult levels over the first 30 days of life. By contrast, in elderly patients the elimination half-life is usually some two to three times longer than in younger adults, any decrease in renal function in the elderly may result in increased half-life. The elimination half-life is only slightly increased in hepatic or renal impairment; renal impairment alone results in increased biliary excretion, hepatic impairment alone results in a compensatory increase in renal elimination. Ceftriaxone crosses non-inflamed and inflamed meninges, resultant concentrations being some 4% - 17% of the corresponding plasma levels. Pharmacodynamics Ceftriaxone is a 2-aminothiazolyl methoxyimino third generation cephalosporin derivative with bactericidal activity against gram negative and gram positive organisms. Ceftriaxone inhibits bacterial cell wall synthesis of actively dividing cells by binding to one or more penicillin binding proteins. These proteins are associated with bacterial cell membrane and play a part in cell wall synthesis. The result is a defective cell wall which is osmotically unstable. It is also hypothesised that cephalosporins decrease the availability of an inhibitor of murein hydrolase, an enzyme involved in cell division. If uninhibited the enzyme can destroy the integrity of the cell wall.

Microbiology:	Ceftriaxone usually shows activity against the following organisms <i>in vitro</i> and in the clinical situation, including strains that produce beta lactamase (penicillinase). <i>Gram positive aerobic bacteria:</i> <i>Staphylococcus aureus</i>, <i>Streptococcus agalactiae</i> (<i>Streptococcus</i> group B), <i>Streptococcus bovis</i>, <i>Streptococcus pneumoniae</i>, <i>Streptococcus pyogenes</i> (<i>Streptococcus</i> group A), <i>Streptococcus viridans</i> It should be noted that methicillin resistant <i>Staphylococcus sp.</i> are resistant to cephalosporins, including ceftriaxone, and the majority of enterococci strains, including <i>Enterococcus faecalis</i>, are resistant. Ceftriaxone is ineffective against <i>Listeria monocytogenes</i>. <i>Gram negative aerobic bacteria:</i> <i>Acinetobacter lwoffii</i> (resistant strains exist), <i>Aeromonas sp.</i>, <i>Alcaligenes sp.</i>, <i>Branhamella catarrhalis</i> (<i>Moraxella catarrhalis</i>), <i>Capnocytophaga sp.</i>, <i>Citrobacter sp.</i>, <i>Enterobacter sp.</i> (resistant strains exist), <i>Escherichia coli</i>, <i>Haemophilus ducreyi</i>, <i>Haemophilus influenzae</i>, <i>H. parainfluenzae</i>, <i>Hafnia alvei</i>, <i>Klebsiella sp.</i>, <i>Moraxella sp.</i>, <i>Morganella morganii</i> (<i>Proteus morganii</i>), <i>Neisseria gonorrhoeae</i>, <i>N. meningitidis</i>, <i>Pasteurella multocida</i>, <i>Plesimonas shigelloides</i>, <i>Proteus mirabilis</i>, <i>Proteus vulgaris</i>, <i>Providencia sp.</i>, <i>Salmonella sp.</i>, <i>Serratia sp.</i>, <i>Shigella sp.</i>, <i>Vibrio sp.</i> and <i>Yersinia sp.</i> <i>Anaerobic Gram positive and Gram negative bacteria:</i> <i>Clostridium sp.</i> with the exception of <i>Clostridium difficile</i>, <i>Fusobacterium sp.</i> with the exception of <i>Fusobacterium mortiferum</i> and <i>F. varium</i>, <i>Peptococcus sp.</i> and <i>Peptostreptococcus sp.</i>
Indications	Infections caused by pathogens sensitive to MEDAXONUM, e.g.: - sepsis - meningitis - abdominal infections (peritonitis, infections of the biliary and gastrointestinal tracts) - infections of the bones, joints, soft tissue, skin and of wounds - infections in patients with impaired defence mechanisms - renal and urinary tract infections - respiratory tract infections, particularly pneumonia and ear, nose and throat infections - genitals infections, including gonorrhoea And perioperative prophylaxis of infections
Contraindications	MEDAXONUM is contraindicated in patients with known hypersensitivity to the cephalosporin class of antibiotics. In patients hypersensitive to penicillin, the possibility of allergic cross-reactions should be borne in mind. Ceftriaxone is contraindicated in neonates (≤ 28 days of age) if they require (or are expected to require) treatment with calcium-containing intravenous solutions, including calcium-containing infusions such as parenteral nutrition, because of the risk of precipitation of ceftriaxone-calcium. Use in neonates with jaundice, hypoalbuminaemia, or whom are acidotic, or have other conditions, including prematurity, in whom impairment of bilirubin binding is likely.
Adverse reactions	The known adverse reactions are listed by body system: <i>Central Nervous System:</i> There are a few reports of headache and dizziness. Nervous system disorders: Frequency 'not known': Encephalopathy* <i>*Reversible encephalopathy has been reported with the use of ceftriaxone, particularly when high doses are administered in patients with renal impairment and additional predisposing factors such as older age, pre-existing central nervous system disorders.</i> <i>Dermatological:</i> Cutaneous reactions including allergic dermatitis, erythema multiforme, oedema, pruritis, urticaria, maculopapular rash and exanthema have occurred. There are isolated reports of more severe cutaneous reactions; erythema multiforme, Lyell's syndrome (toxic epidermal necrolysis) and Stevens-Johnson syndrome. <i>Gastrointestinal:</i> These are the most common adverse effects. The most common one is loose stools or diarrhoea. In the case of profuse, watery and bloody diarrhoea, the possibility of pseudomembranous colitis, due to superinfection by <i>Clostridium difficile</i> should be considered, the appropriate investigations made, and if confirmed the appropriate therapy instituted. There are occasional reports of nausea and/or vomiting, stomatitis and glossitis. <i>Genito-urinary tract:</i> Following ceftriaxone administration there are rare reports of urinary precipitates of calcium ceftriaxone, these are symptomatic and reversible, and patients who are very young, non-mobile, or inadequately hydrated appear to be more at risk. Anuria and renal impairment occurred in isolated instances following this reaction. There are reports of glycosuria, haematuria, and oliguria. Rare reports of fungal superinfection of the genital tract.

Haematological: There are reports of all grades of anaemia, including haemolytic anaemia, agranulocytosis, eosinophilia, leucopenia, neutropenia, and thrombocytopenia. Reports of positive Coomb's test. Rare reports of prolongation of prothrombin time associated with ceftriaxone.

Hepatic: There are reports of transient elevations in liver function tests.

Immunological: There have been rare reports of anaphylactic type reactions, including bronchospasm and angioneurotic oedema. Drug fever and shivering has also been reported rarely.

Injection site: Immediately following intramuscular injection pain or discomfort may be experienced at the injection site, this is usually tolerable and transient. There are rare instances of local phlebitis following intravenous administration, which can be minimised by using a slow intravenous injection of 2 – 4 minutes duration.

Renal: Rare reports of elevated serum creatinine levels. Precipitate of calcium ceftriaxone, sometimes mistaken for gallstones have been detected by ultrasound scans. These biliary precipitates are usually seen following adult dosage in excess of 2g per day, or the equivalent in children. These precipitates may occasionally result in symptoms, in such instances non-surgical management is advocated, and discontinuation of ceftriaxone should be considered. These biliary precipitate usually disappear following therapy cessation. There appears to be an increased risk of such precipitates associated with therapy duration in excess of 14 days, renal failure, total parenteral nutrition or inadequate hydration. Isolated reports of pancreatitis, although definitive causality with ceftriaxone has not been demonstrated.

Superinfection: In common with all broad spectrum antibiotics, superinfection with resistant organisms, yeast or fungi may occur. If these occur, the appropriate therapeutic measures should be instituted.

Precautions and Warnings

The recommended daily dosages should not be exceeded.

Ceftriaxone must not be mixed or administered simultaneously with calcium-containing solutions or products, even via different infusion lines.

Calcium containing solutions or products must not be administered within 48 hours of last administration of ceftriaxone.

Cases of fatal reactions with calcium-ceftriaxone precipitates in lungs and kidneys in both term and premature neonates have been described. In some cases the infusion lines and times of administration and calcium-containing solutions differed.

In patients other than neonates, ceftriaxone and calcium-containing solutions may be administered sequentially to one another if the infusion lines are thoroughly flushed between infusions with a compatible fluid.

Diluents containing calcium, such as Ringer's solution or Hartmann's solution, are not to be used to reconstitute Ceftriaxone vials or to further dilute a reconstituted vial for intravenous administration because a precipitate can form. Ceftriaxone must not be administered simultaneously with calcium-containing intravenous solutions, including continuous calcium-containing infusions such as parenteral nutrition via a Y-site, because precipitation of ceftriaxone-calcium can occur.

Caution should be exercised in administering ceftriaxone to patients who have previously experienced a hypersensitivity reaction, especially anaphylaxis, to penicillins or other non-cephalosporin beta lactam antibiotics. There are occasional reports of cross allergenicity between these antibiotics and cephalosporins. Should anaphylactic shock occur it requires immediate treatment.

Dosage reduction is required in patients with severe renal impairment accompanied by hepatic impairment (see also "Method and Route of Administration").

Due to high plasma binding ceftriaxone can displace bilirubin with a potential of inducing bilirubin encephalopathy and hyperbilirubinaemia. It should therefore be used with caution in neonates, and is contraindicated in acidotic, hypoalbuminaemic or jaundiced neonates (see also "Contraindications"). Great caution is advised in use in premature infants.

Ceftriaxone sodium may precipitate in the gall bladder, such precipitates are detectable as shadow in ultrasound scans. Although this may occur in patients of any age, it is more likely in infants and small children who receive usually a proportionately greater dose on a body weight basis. There is an increased risk of biliary precipitation if dosages in excess of 80 mg / kilogram body weight are used, and therefore such usage should be avoided. In the absence of clear evidence for the development of gallstones or acute cholecystitis in children treated with ceftriaxone, and a conservative management of such precipitates is advocated.

It is characteristic of the cephalosporin antibiotic class to be absorbed onto red blood cell membranes, and react with antibodies produced against the drug resulting in a positive Coomb's test and, on occasion, a mild haemolytic anaemia. There may be cross reactivity with penicillins (see also "Interactions with other medicaments and Other forms of interaction").

There have been rare reports of instances of pancreatitis, possibly due to biliary obstruction, in some ceftriaxone treated patients. In the majority of instances risk factors for biliary stasis and biliary sludge, such as preceding major therapy, severe disease, or on total parenteral nutrition, were present in these patients. However, the possibility of an initiating or cofactor role of ceftriaxone in biliary precipitation in such instances cannot definitively be excluded.

It is recommended that regular and periodic blood count determinations be done during treatment (see also "Adverse reactions").

Although rare, there is an increased risk of reversible symptomatic precipitates of calcium ceftriaxone in urine in patients who are very young, non-mobile, or inadequately hydrated (see also "Adverse Reactions").

Pseudomembranous colitis has been reported with almost all antibiotics, including ceftriaxone, ranging in severity from mild to life-threatening. It is therefore very important to consider this diagnosis in patients who show diarrhoea after they have been administered antibiotics.

Prolonged use of ceftriaxone may result in the overgrowth of non-susceptible organisms and therefore careful observation of the patient is essential. If super infection occurs during treatment, appropriate measures should be taken.

Each gram of ceftriaxone contains approximately 3.6mmol sodium. This should be taken into consideration when treating sodium restricted patients with high dosage regimens.

Drug interactions & Other forms of interactions**Drugs:**

Aminoglycoside antibiotics: During simultaneous administration with ceftriaxone no interference in the action or increase in the nephrotoxicity of the aminoglycosides was observed.

Chloramphenicol: Antagonistic effects have been observed with chloramphenicol and ceftriaxone.

Diuretics: Simultaneous administration of diuretics and ceftriaxone has not led to renal impairment.

Laboratory tests:

Coombs test: Rarely this may result in a false positive result.

Galactosaemia: Ceftriaxone administration may result in a false positive test.

Glucose in urine: Non enzymatic glucose in urine tests may result in a false positive result. Glucose in urine tests during ceftriaxone therapy should be performed with enzymatic test methods.

Food:

Alcohol: Ceftriaxone has an N-methylthiotriazine ring instead of the N-methylthiotetrazole side chain present in other cephalosporins and associated with a disulfiram like reaction in association with alcohol. It may still have the potential to cause such a reaction with alcohol and thus patients should be cautioned to avoid alcohol during ceftriaxone therapy.

Pregnancy and Lactation

Pregnancy: Ceftriaxone was not associated with adverse foetal development in animal studies, but such studies are not always indicative of behaviour in man. There are no adequate and well controlled studies in man and safety in pregnancy has not been established. Use in pregnancy is not recommended unless such use is essential, and consideration should be given to the potential benefit versus the unknown potential hazards.

Lactation: Ceftriaxone is excreted in small amounts in breast milk. Caution is advised in breast feeding and it is suggested that breast feeding be discontinued during therapy.

Effects on Ability to Drive and Use Machines

No known effects

Route of Administration

IM injection

IV injection
IV infusion

**Dosage and
Administration**

Standard dosage

Adults and children over twelve years:

The usual dosage is 1-2g of **MEDAXONUM** administered as a single daily dose. In severe cases or in infections caused by moderately sensitive organisms, the dosage may be increased to 4g, administered once daily.

Neonates, infants and children up to twelve years:

Neonates (up to two weeks): A daily dose of 20-50mg/kg bodyweight, not to exceed 50mg/kg bodyweight, on account of the immaturity of the infant's enzyme systems. It is not necessary to differentiate between premature and infants born at term.

Infants and children (three weeks to twelve years): A daily dose of 20-80mg/kg.

For children with bodyweights of 50kg or more, the usual adult dosage should be used. Intravenous doses of 50mg or more per kg should be given by slow infusion over at least 30 minutes.

Elderly:

The dosages recommended for adults require no modification in the case of geriatric patients.

Duration of therapy: The duration of therapy varies according to the course of the disease. As with antibiotic therapy in general, administration of **MEDAXONUM** should be continued for a minimum of 48 to 72 hours after the patient has become afebrile or evidence of bacterial eradication has been obtained.

Combination therapy: Synergy between Ceftriaxone and aminoglycosides has been demonstrated with many gram-negative bacilli under experimental conditions. Although enhanced activity of such combinations is not always predictable, it should be considered in severe, life-threatening infections due to microorganisms such as *Pseudomonas aeruginosa*. Because of physical incompatibility the two drugs must be administered separately at the recommended dosages.

Special dosage instructions

Meningitis: In bacterial meningitis in infants and children, treatment begins with doses of 100mg/kg (not to exceed 4g) once daily. As soon as the causative organism has been identified and its sensitivity determined, the dosage can be reduced accordingly.

Gonorrhea (acute, uncomplicated): For the treatment of gonorrhea (penicillinase-producing and non-penicillinase-producing strains) an intramuscular administration of a single 250mg dose is recommended. Administration of probenecid simultaneously is not indicated.

Peri-operative prophylaxis: To prevent postoperative infections in contaminated or potentially contaminated surgery, the recommended approach depending on the risk of infection is a single dose of 1-2g **MEDAXONUM** administered 30-90 minutes prior to surgery. In colorectal surgery, concurrent (but separate) administration of **MEDAXONUM** and a 5-nitroimidazole, e.g. ornidazole, has been proven effective.

Impaired renal and hepatic function: In patients with *impaired renal function*, there is no need for dosage adjustment provided *hepatic function* is intact. Only in cases of *preterminal renal failure* (creatinine clearance < 10ml/min) the dosage should not exceed 2 g daily. In patients with *liver damage*, there is no need for the dosage to be reduced provided *renal function* is intact. In cases of concomitant *severe renal and hepatic dysfunction*, the plasma concentrations of ceftriaxone should be determined at regular intervals.

In patients undergoing dialysis no additional supplementary dosing is required following the dialysis. Serum concentrations should be monitored, however, to determine whether dosage adjustments are necessary, since the elimination rate in these patients may be reduced.

Directions for use:

Reconstitution should be carried out under suitable conditions and with appropriate precautions to prevent microbial contamination. It is recommended that freshly prepared solutions be used; the solutions retain satisfactory potency for six hours at 25°C, and up to twenty-four hours at 2°C – 8°C. Following reconstitution, a pale yellow to amber solution is produced depending on the concentration and the length of storage. This characteristic of the active ingredient is of no significance for the efficacy or tolerance of the drug.

Intramuscular injection: 1g should be reconstituted using 3.5ml of 1.0% Lidocaine. Administration is by deep intragluteal injection. If the dosage exceeds 1 g, the dose should be divided equally and administered at more than one injection sites. Solutions reconstituted in Lidocaine should not be administered intravenously.

Intravenous injection: 1g should be reconstituted in 10 ml of water for injection. Administration is by slow intravenous injection over two to four minutes.

Intravenous infusion: 1g of the powder was dissolved in 20 ml respectively of below mentioned calcium free solution:

- Dextrose Injection BP 5%
- Sodium Chloride Injection BP
- Sodium Chloride and Dextrose Injection BP
- Dextran 6% in Dextrose Injection BP 5%,
- Hydroxyethyl Starch 6% infusions

The infusion should be administered over at least 30 minutes

Overdosage and Antidotes	It would not be anticipated that overdosage would be a practical problem, and literature contains no reports of overdose. Symptoms of overdosage would be anticipated to be similar to those described under adverse effects. There is no specific antidote to ceftriaxone sodium, and haemodialysis or peritoneal dialysis would not reduce serum concentrations of the drug. Treatment should be symptomatic and supportive, and close observation should be maintained.
Packing and Pack sizes	MEDAXONUM 1g powder for injection is packed in clear glass (Type I) vials of 10ml nominal capacity, sealed with a grey bromobutyl rubber stopper and an aluminium cap with a plastic flip-off seal. Each vial is labelled, label includes batch number, manufacture and expiry dates. The requisite number of vials (10, 50, 100) are placed in a carton together with the patient information leaflet.
Storage conditions	Powder for injection: Store below 30°C in the original package, in order to be protected from light Reconstituted solution: Recommended immediate use of the reconstituted solution and if necessary the solution may be stored for no more than 6 hours at 25°C and not more than 24 hours at 2°C – 8°C.
Shelf life	24 months
Pharmaceutical Form	Sterile powder for reconstitution for injection
Incompatibilities	Solutions of ceftriaxone sodium should not be added to or mixed with solutions concerning other agents. Specifically ceftriaxone is incompatible with calcium containing solutions.
Manufacturer	Medochemie Ltd – Limassol, CYPRUS
Manufacturing site:	Medochemie (Far East) Ltd – Aseptic Cephalosporin Facility, No. 10, 12 and 16 VSIP II-A, Street 27, Vietnam-Singapore Industrial Park II-A, Vinh Tan Commune, Tan Uyen Town, Binh Duong Province, Vietnam
Product Registration Holder	Komedic Sdn Bhd 4, Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya Selangor, MALAYSIA
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